StackStore

Team & Track: Devil2b | Track 6

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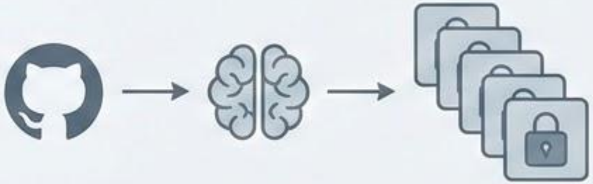
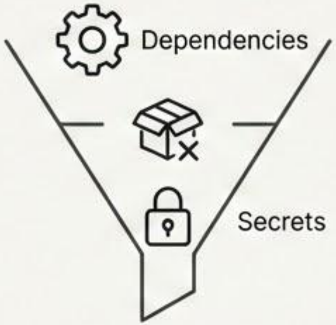
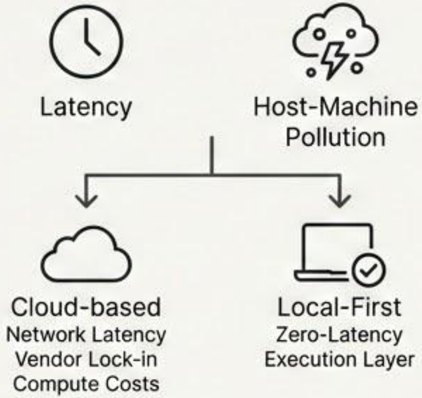
Product Vision

StackStore is an AI-native DevSecOps orchestrator. We instantly translate standard GitHub repositories into securely isolated, zero-config local execution environments.



The Problem





The Configuration Bottleneck

Cloning a repository demands manual resolution of runtime dependencies, OS-level package conflicts, and explicit secret management.

Who is facing this

Software Engineers, Open-Source Contributors, and Enterprise Onboarding teams.

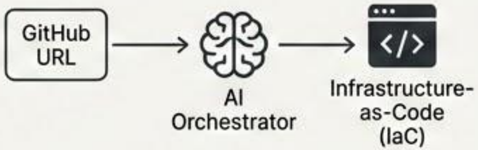
Why it matters

Manual configuration introduces severe onboarding latency and host-machine pollution. Cloud alternatives introduce network latency, enforce vendor lock-in, and incur persistent compute costs. Developers require a zero-latency, local-first execution layer.

Our Solution

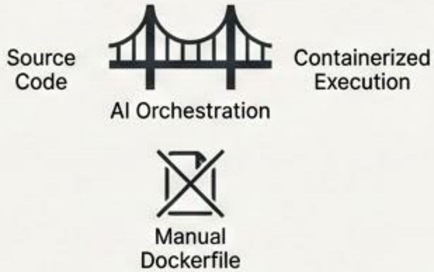
What your product does:

StackStore ingests a standard GitHub URL and leverages an AI Orchestrator to dynamically generate deterministic infrastructure-as-code.



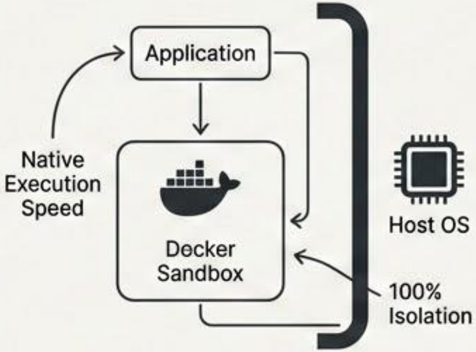
Key idea behind your approach:

We engineered an automated bridge between raw source code and containerized execution, replacing manual Dockerfile configuration with AI orchestration.



How it solves the problem:

It instantly mounts the application into an ephemeral, locally hosted Docker sandbox, guaranteeing native execution speed and 100% isolation from the developer's host OS.



Supported Scope:

Currently optimized to orchestrate modern full-stack architectures, specifically targeting Node.js, Python, and enterprise Java ecosystems.

Node.js (MERN/PERN)

Python (FastAPI/Flask/Data Science)

Enterprise Java (Spring Boot)

How It Works

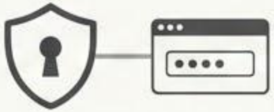
Ingestion

Securely fetches repository manifests via GitHub API—no heavy cloning.

Static Analysis

AI maps architecture to output an executable JSON blueprint.

Just-in-Time Secret Injection



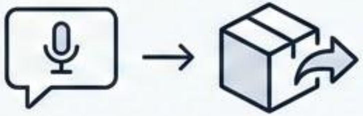
UI intercepts build for secure, just-in-time injection of missing secrets.

Ephemeral Boot

Electron backend pipes a stateless terminal directly into the active Docker container.

Key Features

Conversational Infrastructure Editing



Dynamically mutate container state via natural language. AI safely regenerates blueprints & rebuilds topology without data loss.

'Zero-Assumption' OS Bootstrapping

AI autonomously injects core dependencies before initializing modern package managers, ensuring zero assumptions.

Automated Resource Lifecycle & Smart Resume

Graceful teardowns and fast wakeups. Guarantees zero host memory leaks upon session pause and resume.

Proven Multi-Ecosystem Execution

Benchmarked against complex, diverse public repositories (Java, Python, Node.js) with seamless handling.

Architecture & Tech Stack

Frontend (Client)

React, Vite, and xterm.js for isolated, stateless terminal rendering.

Backend / Bridge

Electron & Node.js (child_process API) handling IPC state management and native Docker daemon telemetry.

AI Services (Orchestrator)

Python / Azure OpenAI (GPT-4o) / Gemini handling unstructured-to-structured deterministic JSON generation.

AI Integration & Enhacements

Where and how AI is used:

Reading unstructured codebase files to output strict, deterministic JSON infrastructure blueprints, and interpreting natural language to dynamically edit them.

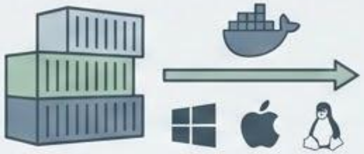
Why AI was used:

Traditional AST parsers cannot deduce implicit environment requirements. Our LLM infers the necessity of specific database connection strings purely by analyzing application routing logic.

Enhancements:

We engineered prompt constraints and temperature controls to force a generative model to act as a strict compiler, outputting rigid JSON arrays that the local engine can parse with a near 0% failure rate.

Scalability & Future Scope



Scalability

Pivoting execution to standard OCI containers (Docker).

Native compatibility & horizontal scaling across Windows, macOS, & Linux out-of-the-box.

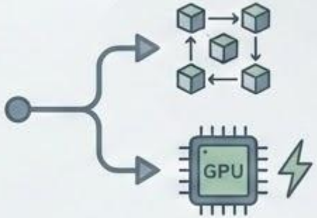
GTM Strategy

Targeting enterprise engineering teams.

Reducing day-one developer onboarding from days to minutes.

Current Challenges

Mitigating LLM token variance. Maintaining absolutely deterministic secret generation generation across multi-layered enterprise monorepos.



Future Roadmap

Upgrading orchestrator for complex docker-compose (microservices).

Enabling direct PCIe passthrough for local ML model training on host GPUs.